VLMEvalKit: An Open-Source Toolkit for Evaluating Large Multi-Modality Models

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Abstract

We present VLMEvalKit: an open-source toolkit for evaluating large multi-modality models based on PyTorch. The toolkit aims to provide a **user-friendly** and **comprehensive** framework for researchers and developers to evaluate existing multi-modality models and publish **reproducible** evaluation results. In VLMEvalKit, we implement over **70 different large multi-modality models**, including both proprietary APIs and open-source models, as well as more than **20 different multi-modal benchmarks**. By implementing a single interface, new models can be easily added to the toolkit, while the toolkit automatically handles the remaining workloads, including data preparation, distributed inference, prediction post-processing, and metric calculation. Although the toolkit is currently mainly used for evaluating large vision-language models, its design is compatible with future updates that incorporate additional modalities, such as audio and video. Based on the evaluation results obtained with the toolkit, we host a comprehensive leaderboard[1] to track the progress of multi-modality learning research. The toolkit is released on GitHub[2] and is actively maintained.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**.

Keywords

open source, multi-modal learning, vision-language model, evaluation, benchmark

[1]https://huggingface.co/spaces/opencompass/open_vlm_leaderboard
[2]https://github.com/open-compass/VLMEvalKit

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MM '24, October 28-November 1, 2024, Melbourne, VIC, Australia
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ACM ISBN 979-8-4007-0686-8/24/10
https://doi.org/10.1145/3664647.3685520

ACM Reference Format:
Haodong Duan, Junming Yang, Yuxuan Qiao, Xinyu Fang, Lin Chen, Yuan Liu, Xiaoyi Dong, Yuhang Zang, Pan Zhang, Jiaqi Wang, Dahua Lin, and Kai Chen. 2024. VLMEvalKit: An Open-Source Toolkit for Evaluating Large Multi-Modality Models. In *Proceedings of the 32nd ACM International Conference on Multimedia (MM '24), October 28-November 1, 2024, Melbourne, VIC, Australia.* ACM, New York, NY, USA, 4 pages. https://doi.org/10.1145/3664647.3685520

1 Introduction

With the rapid development of Large Language Models (LLMs) [1, 20, 25, 26], Large Multi-Modality Models (LMMs) [21, 24] have also experienced significant advancements. LMMs typically take two or more modalities as input. Most of the research has focused on LMMs for image and text [2, 15], but research has also been extended to other modalities, such as audiotext [6], video [4, 12, 23], or point clouds [27, 28]. Furthermore, there exist LMMs that can simultaneously take more than two modalities as inputs, including proprietary APIs and open-source models [10, 21, 24]. Compared to previous multi-modality models, LMMs, empowered by large language models, exhibit enhanced generalization capability and engage with humans in a variety of conversational styles. These models have not only demonstrated remarkable capabilities in multi-modal perception and reasoning tasks but have also spurred a range of innovative applications.

Quantitative evaluation is crucial in the development of LMMs. As general-purpose models, LMMs must undergo rigorous evaluation in a diverse range of tasks and domains. Comprehensive evaluations not only help users discern the strengths and weaknesses of an LMM, but also offer valuable feedback to developers for ongoing refinement. Unlike 'pre-GPT' models, evaluating LMMs on diversified quantitative benchmarks has become a common practice, both in academic works [14, 29] and commercial APIs [21, 24].

Despite the importance of evaluating LMMs, conducting assessments across dozens of benchmarks can be a daunting task, particularly for small research teams. One must prepare data based on numerous repositories and manage potential environmental conflicts. Moreover, the authors of benchmarks may not provide

Benchmark	Keyword	Benchmark	Keyword
MMBench	All-round Capability	CCBench	Chinese Culture
MME	All-round Capability	SEEDBench	All-round Capability
SEEDBench2-Plus	Text-Rich Comprehension	MMT-Bench	Multi-Task
MMMU	Multi-Modal Examination	MMStar	Data Contamination
MathVista	Multi-Modal Math	COCO Caption	Image Caption
OCRBench	Text Understanding	TextVQA	Text Recognition
AI2D	Diagram Understanding	ChartQA	Chart Understanding
DocVQA	Document Understanding	RealWorldQA	Driving Scenarios
HallusionBench	Hallucination & Illusion	POPE	Object Hallucination
MMBench-Video	Video Understanding	OCRVQA	Text Recognition
MLLMGuard	Multi-Modal Safety	AesBench	Aesthetics
MMVet	Subjective Evaluation	LLaVABench	Subjective Evaluation

Table 1: VLMEvalKit supports the evaluation of more than 20 different benchmarks, covering diversified tasks, topics, and scenarios. In the table, we list some benchmarks and the corresponding keywords.

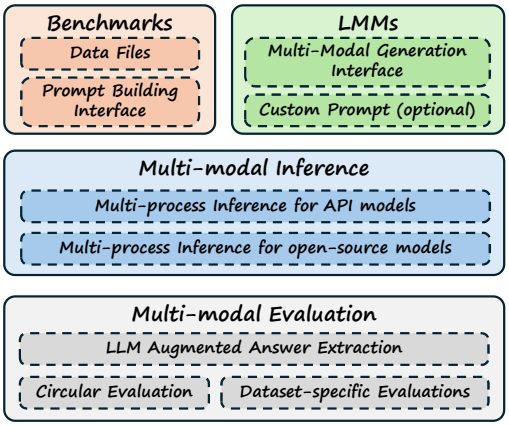

Figure 1: Major Components in VLMEvalKit.

evaluation results for all LMMs in which users are interested, thus requiring significant effort to compile uncompleted results. To alleviate this challenge, we developed VLMEvalKit, an open-source toolkit designed to facilitate the evaluation of LMMs.

VLMEvalKit aims to provide a comprehensive, user-friendly evaluation framework for researchers and developers to assess existing Large Multi-Modality Models. Currently, the codebase supports more than 70 different large multi-modality models, spaning proprietary APIs and open-source models, and over 20 multi-modal benchmarks covering a wide range of tasks and scenarios (see Tab. 1). The codebase's straightforward design simplifies the integration of new benchmarks or LMMs. Typically, a developer only needs to prepare a single data file or implement a single interface to support new benchmarks or LMMs. Beyond its extensive collection and simplified design, VLMEvalKit significantly eases the work of comprehensive evaluation. Users can launch evaluations across multiple supported LMMs and benchmarks with a single command, generating well-structured evaluation results. The entire process eliminates the need for manual data preparation or post-processing, ensuring a seamless and efficient evaluation experience.

To align with real use cases, VLMEvalKit employs generation-based evaluation across all LMMs and benchmarks. Given their general-purpose nature, LMMs may struggle with adhering to specific formats for solving multiple-choice questions. For a fair comparison, VLMEvalKit utilizes large language models as choice extractors when exact matching fails, thereby mitigating the impact of response styles and enhancing the reliability of evaluation, Leveraging the reliable and reproducible evaluation results provided by VLMEvalKit, we maintain a comprehensive leaderboard to monitor the advancement of LMM development. The leaderboard disseminates valuable insights to the community and has garnered widespread recognition from both academia and industry.

VLMEvalKit is publicly available at https://github.com/open-compass/VLMEvalKit under the Apache 2.0 License. The repository includes the complete source codes along with detailed instructions for installation, evaluation, and further development. In subsequent sections, we will delve into the design and features of VLMEvalKit, present and analyze the evaluation results obtained, and discuss potential future developments.

2 VLMEvalKit: Design & Features

Fig. 1 displays the major components in VLMEvalKit. In this section, we will describe each single component in detail.

Benchmarks. VLMEvalKit encompasses dozens of multi-modal benchmarks that span a broad spectrum of tasks and scenarios, as detailed in Tab. 1. The benchmarks are first processed into `.tsv` files, where evaluation samples are stored as individual lines. Each evaluation sample typically includes the `index`, `question`, `answer`, `image` (one or more images encoded in `base64`), and `choices` for multi-choice questions. Each dataset class supports a `.build_prompt()` interface, which the evaluation script utilizes to construct evaluation samples into multi-modal messages. A multi-modal message comprises an interleaved sequence of contents across various modalities. For illustration, consider the following example:

```
[
    dict(type='image', value='path or url of the image'),
    dict(type='image', value='path or url of the image'),
    dict(type='text', value='Please list all the objects that
appear in the above images.')
]
```

A multi-modal message primarily incorporates `image` and `text` modalities. The format is currently used for evaluating image [16, 31] and video [9] understanding benchmarks. It is also extensible and can accommodate additional modalities such as audio or point clouds.

LMMs. VLMEvalKit supports over 70 LMMs, including both commercial APIs and open-source models. A unified `.generate()` interface has been implemented for all LMMs, which accepts a multi-modal message as input and returns the response string. For LMMs that are limited to processing a single image-text pair, the concatenated text messages and the first image are adopted as the default input. To provide flexibility for LMM developers, the `.generate()` interface also includes `dataset_name` as an additional argument. An LMM can optionally implement its own `.build_prompt()` interface to construct custom multi-modal messages, and determine whether to utilize custom multi-modal messages based on the `dataset_name` flag. The unified interface eases the process of comprehensive evaluation. Each newly supported LMM / benchmark can be directly evaluated with all existing benchmarks / LMMs.

Multi-modal Inference. To expedite the multi-modal inference, we support parallelized inference for both commercial APIs, leveraging Python's `multiprocessing`, and open-source models, which are distributed in parallel across multiple GPUs. Additionally, we have implemented a robust inference process so that an interrupted inference process can be resumed with minimal costs of repeated calculation or API calls.

Multi-modal Evaluation. Predictions from the LMMs will be evaluated based on the specific question format to derive the final metrics. Benchmarks supported in VLMEvalKit can be categorized into three primary types: 1. Multi-choice questions (MCQ), where the model must select from given options and respond with the corresponding label (*e.g.*, A, B); 2. Yes-or-No questions (Y/N), requiring a straightforward 'Yes' or 'No' answer; 3. Open-ended questions, which necessitate a free-form response. Notably, a significant number of LMMs struggle to adhere to the instructions precisely, often producing responses that are not well-formatted for MCQ and Y/N benchmarks. To improve the precision of evaluations and counteract the influence of varied response styles, VLMEvalKit offers the option to integrate LLM-augmented answer extraction specifically for MCQ and Y/N benchmarks. We first adopt exact matching to match the response with the option labels or contents. Should this step fail, the toolkit then prompts an LLM (such as ChatGPT) to match the response with the option that most closely aligns with semantic meaning. This strategy helps us better understand the real performance of LMMs, particularly for commercial APIs [16].

Another notable challenge in assessing MCQ benchmarks is the inherent **variance**. When employing random guessing, an LMM may correctly answer $1/N$ questions for N-option multi-choice benchmarks. Besides, we find that the outcome of an LMM can be significantly influenced by the order of options. These factors make the evaluation results highly variable and the performance gap between LMMs less discernible. To address this, VLMEvalKit offers an option to evaluate all MCQ benchmarks in **Circular** mode: N options of a MCQ will be shifted in circular N times to formulate N new questions. The results count only if an LMM accurately answers all N circular-shifted MCQs. The CircularEval strategy can more effectively assess the real comprehension of an LMM on MCQ benchmarks, allowing users to identify more pronounced performance disparities between models.

For open-ended benchmarks, VLMEvalKit follows the original practice for conducting the evaluation. Subjective benchmarks like MMVet [30] or LLaVABench [15] adopts GPT-4 for marking, based on the semantic similarity between LMM responses and the reference answer. For VQA benchmarks [19, 22], we follow the standard practices to calculate accuracies based on heuristic matching.

3 Evaluation Results

Utilizing VLMEvalKit, one can conduct comprehensive evaluations of an LMM across numerous benchmarks to gain a thorough understanding of its strengths and weaknesses. We publish all evaluation results on OpenVLM Leaderboard. Our core leaderboard is based on evaluations from eight distinct benchmarks: 1. MMBench v1.1 [test] [16][3] (all-round capability); 2. MMStar [3] (data contamination); 3. MMMU [31] [val] (multi-modal examination); 4. MathVista [mini-test] [18] (multi-modal math); 5. HallusionBench [13] (hallucination & illusion); 6. AI2D [test] [11] (diagram understanding); 7. OCRBench [17] (text understanding); 8. MMVet [30] (subjective evaluation). The selection encompasses a diverse array of tasks, and the average score across these benchmarks serves as a reliable indicator of the general capabilities of LMMs. In Tab. 2, we present the performance of top-10 commercial APIs and the top-10 open-source models evaluated on this suite of benchmarks.

Commercial APIs. Generally, commercial APIs exhibit a significant performance advantage over open-source LMMs. Notably, the top-5 performing LMMs are all proprietary APIs, with GPT-4o surpassing the highest-ranked open-source model, InternVL-Chat-V1.5, by over 8 points in average score. API models particularly excel on two benchmarks: MMMU [31] and MMVet [30]. On MMMU, four API models achieve above 60% accuracy, with an additional two surpassing 50%, whereas all open-source LMMs fall below the 50% threshold. Similarly, on MMVet, seven of the top-10 API models score above 60%, outperforming all open-source counterparts. Although the exact reasons for this disparity cannot be definitively confirmed, it is plausible that proprietary API LMMs leverage larger language encoders (at the 100B scale, *etc.*), which may enhance their knowledge level and better align with human preferences. We also note that the true capabilities of proprietary LMMs might still be understated due to content moderation policies employed, which can lead to the rejection of certain question types. Quantitative analysis indicates that the policies result in at least 2% and 4% reduction in average scores for GPT-4o and Gemini-1.5-Pro, respectively.

Open-source LMMs. The top-10 open-source LMMs predominantly utilize small to medium-scale language encoders, with parameter sizes ranging from 4B to 34B. InternVL-Chat-V1.5 [5] stands out as the highest performer among all open-source LMMs, surpassing the 2_{nd}-best model by 2 points in both average score and on the comprehensive MMBench benchmark. Leveraging the largest language encoder, LLaVA-NeXT-Yi-34B [14] achieves the top score on MMMU, a benchmark that demands extensive knowledge. InternLM-XComposer2 [7] impresses with its capabilities in multi-modal math and diagram understanding, while two LMMs based on GLM [8] notably excel on OCRBench and MMVet. Additionally, the lightweight Mini-InternVL, featuring a 4B parameter size, demonstrates performance on par with the early versions of GPT-4v and GeminiPro, highlighting its efficiency and effectiveness.

[3]We report the average score of English and Chinese test splits.

Acknowledgement

This project is supported by the National Key R&D Program of China (No.2022ZD0161600), Shanghai Postdoctoral Excellence Program (No.2023023), China Postdoctoral Science Fund (No.2024M751559), and Shanghai Artificial Intelligence Laboratory.

References

[1] Anthropic. 2024. The claude 3 model family: Opus, sonnet, haiku. (2024).
[2] Lin Chen, Jisong Li, Xiaoyi Dong, Pan Zhang, Conghui He, Jiaqi Wang, Feng Zhao, and Dahua Lin. 2023. ShareGPT4V: Improving Large Multi-Modal Models with Better Captions. *arXiv:2311.12793* (2023).
[3] Lin Chen, Jinsong Li, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Zehui Chen, Haodong Duan, Jiaqi Wang, et al. 2024. Are We on the Right Way for Evaluating Large Vision-Language Models? *arXiv:2403.20330* (2024).
[4] Lin Chen, Xilin Wei, Jinsong Li, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Zehui Chen, Haodong Duan, Bin Lin, Zhenyu Tang, et al. 2024. Sharegpt4video: Improving video understanding and generation with better captions. *arXiv preprint*

Method	Param.	Avg. Score	MMBench	MMStar	MMMU	Math.	Hallu.	AI2D	OCR.	MMVet
GPT-4o-20240513	N/A	**69.9**	**82.2**	**63.9**	**69.2**	61.3	**55.0**	**84.6**	736	**69.1**
Claude3.5-Sonnet	N/A	67.9	78.5	62.2	65.9	**61.6**	49.9	80.2	788	66.0
CongRong	N/A	65.5	80.7	60.6	48.3	61.0	50.6	82.4	**827**	57.5
Gemini-1.5-Pro	N/A	64.4	73.9	59.1	60.6	57.7	45.6	79.1	754	64.0
GPT-4v-20240409	N/A	63.5	79.8	56.0	61.7	54.7	43.9	78.6	656	67.5
InternVL-Chat-V1.5	25.5B	**61.7**	**79.7**	**57.1**	46.8	54.7	47.4	80.6	720	55.4
GLM-4v	N/A	60.8	78.6	53.2	45.6	45.6	44.9	76.1	814	60.7
Step-1v	N/A	59.5	78.0	50.0	49.9	44.8	48.4	79.2	625	63.3
GLM-4v-9B	13.9B	59.1	67.9	54.8	46.9	51.1	45.0	71.2	**776**	**58.0**
MiniCPM-Llama3-V2.5	8.5B	58.8	72.0	51.8	45.8	54.3	42.4	78.4	725	52.8
InternLM-XComposer2-VL-4KHD	8.6B	58.8	76.5	55.3	39.7	59.4	42.5	81.0	675	48.2
WeMM	9.6B	58.3	75.7	57.0	45.3	54.9	**47.5**	77.9	628	45.0
Qwen-VL-Max	N/A	58.3	74.6	49.5	52.0	43.4	41.2	75.7	684	61.8
InternLM-XComposer2-VL	8.7B	57.1	77.6	56.2	41.4	**59.5**	41.0	**81.2**	532	46.7
GPT-4v-20231106	N/A	57.1	73.8	49.7	53.8	48.7	46.5	75.9	516	56.8
CogVLM2-19B-Chat	19.5B	56.3	70.7	50.5	42.6	38.6	41.3	73.4	757	57.8
Mini-InternVL-Chat-4B-V1.5	4.1B	56.2	69.7	53.1	45.1	54.6	43.0	77.0	639	43.6
Gemini-1.0-Pro	N/A	56.1	69.7	38.6	49.0	46.5	45.7	72.9	680	58.6
LLaVA-NeXT-Yi-34B	34.8B	55.0	77.8	51.6	**48.8**	40.4	34.8	78.9	574	50.7
Phi-3-Vision	4.1B	53.6	65.2	47.7	46.1	44.6	39.0	78.4	637	44.1

Table 2: The evaluation results of LMMs on selected benchmarks. The table displays the top-10 commercial APIs and top-10 open-source LMMs (till 2024.06.30), in the descending order of average score. When calculating the average score, scores of each benchmark are normalized to the range of 0 to 100. When reporting the parameter size, '1B' means 10^9 parameters. Commercial APIs are denoted with blue background. Bold and underline indicates the best and second-best performance in each group.

arXiv:2406.04325 (2024).

[5] Zhe Chen, Weiyun Wang, Hao Tian, Shenglong Ye, Zhangwei Gao, Erfei Cui, Wenwen Tong, Kongzhi Hu, Jiapeng Luo, Zheng Ma, Ji Ma, Jiaqi Wang, Xiaoyi Dong, Hang Yan, et al. 2024. How Far Are We to GPT-4V? Closing the Gap to Commercial Multimodal Models with Open-Source Suites. arXiv:2404.16821 [cs.CV]

[6] Yunfei Chu, Jin Xu, Xiaohuan Zhou, Qian Yang, Shiliang Zhang, Zhijie Yan, et al. 2023. Qwen-audio: Advancing universal audio understanding via unified large-scale audio-language models. *arXiv:2311.07919* (2023).

[7] Xiaoyi Dong, Pan Zhang, Yuhang Zang, Yuhang Cao, Bin Wang, Linke Ouyang, Xilin Wei, Songyang Zhang, Haodong Duan, Maosong Cao, Wenwei Zhang, et al. 2024. InternLM-XComposer2: Mastering Free-form Text-Image Composition and Comprehension in Vision-Language Large Model. *arXiv:2401.16420* (2024).

[8] Zhengxiao Du, Yujie Qian, Xiao Liu, Ming Ding, Jiezhong Qiu, Zhilin Yang, and Jie Tang. 2022. GLM: General Language Model Pretraining with Autoregressive Blank Infilling. In *ACL*. 320–335.

[9] Xinyu Fang, Kangrui Mao, Haodong Duan, Xiangyu Zhao, Yining Li, Dahua Lin, and Kai Chen. 2024. MMBench-Video: A Long-Form Multi-Shot Benchmark for Holistic Video Understanding. *arXiv:2406.14515* (2024).

[10] Jiaming Han, Kaixiong Gong, Yiyuan Zhang, Jiaqi Wang, Kaipeng Zhang, Dahua Lin, Yu Qiao, Peng Gao, and Xiangyu Yue. 2024. Onellm: One framework to align all modalities with language. In *CVPR*. 26584–26595.

[11] Aniruddha Kembhavi, Mike Salvato, Eric Kolve, Minjoon Seo, Hannaneh Hajishirzi, and Ali Farhadi. 2016. A diagram is worth a dozen images. In *ECCV*. Springer, 235–251.

[12] Kunchang Li, Yali Wang, Yinan He, Yizhuo Li, Yi Wang, Yi Liu, Zun Wang, Jilan Xu, Guo Chen, Ping Luo, et al. 2024. Mvbench: A comprehensive multi-modal video understanding benchmark. In *CVPR*. 22195–22206.

[13] Fuxiao Liu, Tianrui Guan, Zongxia Li, Lichang Chen, Yaser Yacoob, et al. 2023. Hallusionbench: You see what you think? or you think what you see? an image-context reasoning benchmark challenging for gpt-4v (ision), llava-1.5, and other multi-modality models. *arXiv:2310.14566* (2023).

[14] Haotian Liu, Chunyuan Li, Yuheng Li, Bo Li, Yuanhan Zhang, Sheng Shen, and Yong Jae Lee. 2024. LLaVA-NeXT: Improved reasoning, OCR, and world knowledge. https://llava-vl.github.io/blog/2024-01-30-llava-next/

[15] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. 2023. Visual instruction tuning. *arXiv:2304.08485* (2023).

[16] Yuan Liu, Haodong Duan, Yuanhan Zhang, Bo Li, Songyang Zhang, Wangbo Zhao, Yike Yuan, Jiaqi Wang, Conghui He, et al. 2023. MMBench: Is your multi-modal model an all-around player? *arXiv:2307.06281* (2023).

[17] Yuliang Liu, Zhang Li, Hongliang Li, Wenwen Yu, Mingxin Huang, Dezhi Peng, Mingyu Liu, Mingrui Chen, Chunyuan Li, Lianwen Jin, et al. 2023. On the hidden mystery of ocr in large multimodal models. *arXiv:2305.07895* (2023).

[18] Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chunyuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-Wei Chang, Michel Galley, and Jianfeng Gao. 2023. Mathvista: Evaluating mathematical reasoning of foundation models in visual contexts. *arXiv:2310.02255* (2023).

[19] Anand Mishra, Shashank Shekhar, Ajeet Kumar Singh, and Anirban Chakraborty. 2019. Ocr-vqa: Visual question answering by reading text in images. In *ICDAR*. IEEE, 947–952.

[20] OpenAI. 2023. ChatGPT. https://openai.com/blog/chatgpt.

[21] OpenAI. 2023. GPT-4 Technical Report. *ArXiv* abs/2303.08774 (2023).

[22] Amanpreet Singh, Vivek Natarajan, Meet Shah, Yu Jiang, Xinlei Chen, Dhruv Batra, Devi Parikh, and Marcus Rohrbach. 2019. Towards vqa models that can read. In *CVPR*. 8317–8326.

[23] Enxin Song, Wenhao Chai, Guanhong Wang, Yucheng Zhang, Haoyang Zhou, Feiyang Wu, Haozhe Chi, Xun Guo, et al. 2024. Moviechat: From dense token to sparse memory for long video understanding. In *CVPR*. 18221–18232.

[24] Gemini Team, Rohan Anil, Sebastian Borgeaud, Yonghui Wu, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, et al. 2023. Gemini: a family of highly capable multimodal models. *arXiv:2312.11805* (2023).

[25] InternLM Team. 2023. InternLM: A Multilingual Language Model with Progressively Enhanced Capabilities. https://github.com/InternLM/InternLM-techreport.

[26] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. 2023. Llama: Open and efficient foundation language models. *arXiv:2302.13971* (2023).

[27] Runsen Xu, Xiaolong Wang, Tai Wang, Yilun Chen, Jiangmiao Pang, and Dahua Lin. 2023. Pointllm: Empowering large language models to understand point clouds. *arXiv:2308.16911* (2023).

[28] Le Xue, Ning Yu, Shu Zhang, Artemis Panagopoulou, Junnan Li, Roberto Martín-Martín, Jiajun Wu, Caiming Xiong, Ran Xu, et al. 2024. Ulip-2: Towards scalable multimodal pre-training for 3d understanding. In *CVPR*. 27091–27101.

[29] Qinghao Ye, Haiyang Xu, Jiabo Ye, Ming Yan, Anwen Hu, Haowei Liu, Qi Qian, Ji Zhang, et al. 2023. mPLUG-Owl2: Revolutionizing Multi-modal Large Language Model with Modality Collaboration. arXiv:2311.04257 [cs.CL]

[30] Weihao Yu, Zhengyuan Yang, Linjie Li, Jianfeng Wang, Kevin Lin, Zicheng Liu, Xinchao Wang, and Lijuan Wang. 2023. Mm-vet: Evaluating large multimodal models for integrated capabilities. *arXiv:2308.02490* (2023).

[31] Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, et al. 2023. Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. *arXiv:2311.16502* (2023).